from HV-DH1 S1-Q1A (SLD-002 sh 1)

S1-Q1A trip
VCB 630 A 25 kA
CT 150/1 5P20 10 VA
Ith ≥ 25 kA 1 s

R-TX
50/51 Is 100 A, IEC EI, TMS 1.0
50 I≫ 1 500 A, 0.05 s
50N/51N · 86 lock-out

3×1×95 Al 12/20 kV

SA 24 kV

T-A
2 500 kVA 20/0.42 kV
Dyn11 · uk 6 % · cast resin
Pk 19 kW · P0 3.1 kW

BTS 4 000 A ≈ 10 m

CT 4000/1 → Wh

Wh

Q-UI ACB 4 000 A
Ir 3 600 A · ARMS

IL-3 ATS: Q-UI / Q-GI open transition only

MSB-A 4 000 A · Icw 65 kA 1 s · Ipk ≥ 143 kA

G
3~

G-A
2 750 kVA / 2 200 kWe DCC
Isc ≥ 300 % for 2 s
HVO100, 24 h

CT 4000/1 → 51V, Wh

BTS 4 000 A ≈ 40 m

Q-GI ACB 4 000 A
Isd ≤ 2.5 × Ir

I″k 52.3 kA / ip 122 kA

R-ARC
50ARC light + I
trip Q-UI, Q-GI
intertrip S1-Q1A

SPD T1+2

Q-R1 2 000 A	→ UPS-A1 rectifier (sh 2)	
Q-R2 2 000 A	→ UPS-A2 rectifier (sh 2)	
Q-SB 3 200 A	→ BTS 3 200 A ≈ 25 m → SBS-A (sh 2)	
Q-MB 3 200 A	→ MBB-A (sh 2)	
Q-LB 2 500 A	→ load-bank connection	
Q-AUX 250 A	→ block aux. lighting, BMS	

LV TRIP SETTINGS (CALC-003)

Device	Ir	tr	Isd	tsd	Other
Q-UI incomer	3 600 A	3 s	4 × Ir	0.3 s	ZSI, ARMS
Q-SB (SBS)	2 900 A	1 s	2.5 × Ir	0.2 s	ZSI
Q-GI genset	4 000 A	12 s	2.5 × Ir	0.4 s	—
Busway MCCB	800 A	—	4 × Ir	0.1 s	Ii 12 × Ir

NOTES
1. ATS: break-before-make, no closed transition (closed: 81.8 kA / ip 187 kA > MSB rating).
2. ZSI Q-UI ↔ outgoing ACBs; ARMS on Q-UI.
3. Arc energy MSB line side 96 → 3.2 cal/cm², bus 16.8 → 2.4–4.8 cal/cm² (IEEE 1584).
4. Ratings are [ASSUMED] concept values.

P2	2026-10-25	CALC-002/3/4 ratings, IEC symbols, ≥2.5 mm text
P1	2026-10-24	First concept issue

Nordic Hyperscale DC Campus (fictional) · 30 MW IT · SE3
Portfolio D. Wu · public standards and [ASSUMED] data only

Power block A — sheet 1

20 kV feeder, T-A, genset G-A, MSB-A (typical of 24)

CONCEPT · NOT FOR CONSTRUCTION · scale NTS · A3

Drawing no.	Sheet	Rev
DWU-DC-E-SLD-003	**1 / 2**	**P2**